Light

GCSE Computer Science - Memory and Storage (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Memory And Storage

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does RAM stand for?

- **A.** Random Access Memory
- **B.** Read And Modify
- **C.** Readily Available Memory
- **D.** Run Application Memory

Q2 (1 mark)

What does ROM stand for?

- **A.** Run Only Memory

- B. Random Output Memory
- C. Read Only Memory
- D. Rewritable Optical Memory

Q3 (1 mark)

What does it mean for memory to be volatile?

- A. The memory is dangerous and may explode
- B. The memory operates at very high temperatures
- C. The memory can only store temporary files
- D. The memory loses its contents when power is switched off

Q4 (1 mark)

Which of the following is an example of non-volatile storage?

- A. Hard disk drive
- B. CPU cache
- C. RAM
- D. Registers

Q5 (1 mark)

What is the main purpose of RAM in a computer system?

- ○ **A.** To permanently store the operating system
- ○ **B.** To hold data and programs that are currently in use
- ○ **C.** To start up the computer when power is first applied
- ○ **D.** To cool down the processor during intensive tasks

Q6 (1 mark)

What type of instructions are typically stored in ROM?

- ○ **A.** User documents and files
- ○ **B.** Temporary calculation results
- ○ **C.** Internet browser history
- ○ **D.** The BIOS/boot instructions needed to start the computer

Q7 (1 mark)

Why is secondary storage necessary in a computer system?

- ○ **A.** To make the computer run faster
- ○ **B.** To provide permanent storage for data and programs when power is off
- ○ **C.** To replace RAM when it becomes full
- ○ **D.** To connect to the internet

Q8 (1 mark)

What technology do solid state drives (SSDs) use to store data?

- ○ **A.** Electronic circuits with no moving parts
- ○ **B.** Laser beams on reflective discs
- ○ **C.** Spinning magnetic platters
- ○ **D.** Magnetic tape wound on spools

Q9 (1 mark)

How does a magnetic hard disk drive (HDD) store data?

- ○ **A.** By burning microscopic pits into a disc surface
- ○ **B.** By magnetising tiny areas on spinning platters
- ○ **C.** By trapping electrons in silicon chips
- ○ **D.** By using radio waves to encode information

Q10 (1 mark)

What is cloud storage?

- ○ **A.** Storage that floats in the atmosphere
- ○ **B.** A type of RAM that works without electricity
- ○ **C.** Storage built into weather monitoring equipment

- ○ **D.** Data stored on remote servers accessed via the internet

Q11 (1 mark)

What is the purpose of cache memory?

- ○ **A.** To store data permanently when the computer is turned off
- ○ **B.** To hold frequently used data for faster access by the CPU
- ○ **C.** To provide backup copies of important files
- ○ **D.** To display images on the monitor

Q12 (1 mark)

Where are registers located in a computer system?

- ○ **A.** On the motherboard next to the RAM slots
- ○ **B.** Inside the hard disk drive
- ○ **C.** Within the CPU itself
- ○ **D.** In the power supply unit

Q13 (2 marks)

A computer needs both RAM and ROM. Which statement correctly explains why both are necessary?

- ○ **A.** RAM provides permanent storage while ROM provides

temporary storage

○ **B.** RAM stores the startup instructions while ROM stores running programs

○ **C.** RAM holds currently running programs and data, while ROM stores permanent startup instructions that must survive power loss

○ **D.** RAM and ROM are identical and can be used interchangeably

Q14 (2 marks)

Which statement correctly describes an advantage of SSDs over traditional HDDs?

○ **A.** SSDs have larger storage capacities for the same price

○ **B.** SSDs are faster because they have no moving parts, allowing quicker data access

○ **C.** SSDs use less electricity because they spin at lower speeds

○ **D.** SSDs are more reliable because magnetic storage never fails

Q15 (2 marks)

A video production company needs to store 50 terabytes of archived footage. Why might they choose HDDs over SSDs for this purpose?

○ **A.** HDDs offer significantly more storage capacity per pound, making large-scale storage more affordable

○ **B.** HDDs are faster at reading video files

○ **C.** HDDs are more portable and shock-resistant

- ○ **D.** HDDs consume less power during continuous operation

Q16 (2 marks)

What is a key advantage of cloud storage over local storage?

- ○ **A.** Cloud storage is always faster than local storage
- ○ **B.** Cloud storage does not require any internet connection
- ○ **C.** Files can be accessed from any device with internet access, enabling collaboration and remote working
- ○ **D.** Cloud storage is completely free with unlimited capacity

Q17 (2 marks)

A business is considering moving all their confidential financial records to cloud storage. What is a significant concern they should consider?

- ○ **A.** Cloud storage cannot handle financial data formats
- ○ **B.** Cloud servers are always slower than local servers
- ○ **C.** Cloud storage automatically deletes old files
- ○ **D.** Security and privacy risks, as data is stored on third-party servers and transmitted over the internet

Q18 (2 marks)

Place these storage types in order from fastest to slowest access speed: RAM, Register, HDD, Cache

- ○ **A.** HDD, RAM, Cache, Register
- ○ **B.** Cache, Register, HDD, RAM
- ○ **C.** RAM, Register, HDD, Cache
- ○ **D.** Register, Cache, RAM, HDD

Q19 (2 marks)

A laptop suddenly loses power while a user is editing a document. Which components will retain their data?

- ○ **A.** RAM and cache only
- ○ **B.** Registers and RAM only
- ○ **C.** The SSD (or HDD) and ROM only
- ○ **D.** All components retain data because modern computers have backup batteries

Q20 (2 marks)

Why are SSDs generally more reliable than HDDs for use in laptops that are frequently transported?

- ○ **A.** SSDs have no moving parts, making them resistant to damage from physical shock and vibration
- ○ **B.** SSDs are waterproof while HDDs are not
- ○ **C.** SSDs store data in multiple locations automatically
- ○ **D.** SSDs are made of stronger materials than HDDs

Q21 (2 marks)

What is the key difference between main memory and secondary storage in terms of CPU access?

- ○ **A.** There is no difference; the CPU accesses both in the same way
- ○ **B.** Main memory can be accessed directly by the CPU, while secondary storage requires data to be loaded into memory first
- ○ **C.** Secondary storage is accessed directly, while main memory is only used for display
- ○ **D.** Main memory is accessed over the internet, while secondary storage is local

Q22 (2 marks)

How does increasing the amount of cache memory improve computer performance?

- ○ **A.** It reduces the number of times the CPU must access slower RAM, speeding up data retrieval
- ○ **B.** It allows more programs to be saved permanently
- ○ **C.** It increases the clock speed of the processor
- ○ **D.** It provides more space for the operating system files

Q23 (2 marks)

When a computer runs low on RAM, the operating system may use a section of the hard drive as 'virtual memory'. What is the disadvantage of this approach?

- ○ **A.** Virtual memory is permanent and cannot be deleted

- ○ **B.** Virtual memory makes the computer run faster
- ○ **C.** Accessing virtual memory is much slower than RAM because it uses secondary storage
- ○ **D.** Virtual memory uses more electricity than RAM

Q24 (3 marks)

A photographer needs storage for their work. They require: fast access when editing high-resolution images, portability between home and studio, and protection against data loss if the drive is dropped. Which storage solution best meets all these requirements?

- ○ **A.** An external HDD because it offers the largest capacity
- ○ **B.** A portable SSD because it provides fast access speeds, easy portability, and resistance to physical shock due to no moving parts
- ○ **C.** Cloud storage only because it is accessible everywhere
- ○ **D.** An internal HDD because it is the most reliable option

Q25 (3 marks)

Explain the sequence of memory and storage involvement when a user turns on their computer and opens a word processor to edit a document.

- ○ **A.** The document loads directly from the HDD into registers, bypassing all other memory
- ○ **B.** Cache stores the operating system permanently, and registers hold the complete document
- ○ **C.** RAM stores the BIOS, which loads the document directly for editing

○ **D.** ROM provides startup instructions, the operating system loads from SSD/HDD into RAM, the word processor loads into RAM, and the document is loaded into RAM for editing with cache and registers used during active processing

Q26 (3 marks)

A small business is deciding between local server storage and cloud storage for their company files. The business has 10 employees who often work remotely and need to collaborate on documents. They also have concerns about data security and costs. Which analysis is most accurate?

○ **A.** Local storage is always cheaper and more secure than cloud storage

○ **B.** Cloud storage offers easier remote access and collaboration capabilities, though it requires trusting a third-party provider with data security; costs may be lower initially but scale with usage, while local storage requires upfront hardware investment but offers direct control over security

○ **C.** Cloud storage is always faster than local storage for all tasks

○ **D.** There is no difference between cloud and local storage for business use

Q27 (3 marks)

A computer currently has 4GB of RAM and frequently shows slowdowns when multiple applications are open. The user is considering upgrading to 16GB. Explain why this upgrade would likely improve performance.

○ **A.** More RAM increases the CPU clock speed, making everything faster

○ **B.** More RAM would not help because RAM has no effect on multitasking

○ **C.** More RAM allows more applications and data to be held in fast memory simultaneously, reducing reliance on slow virtual memory and enabling smoother multitasking

○ **D.** The improvement would only occur if the hard drive is also replaced

Q28 (3 marks)

A school is planning storage for 500 students, each requiring 2GB of personal file storage plus access to 50GB of shared educational resources. The IT department is comparing options. What total storage capacity is needed, and which solution might be most appropriate?

○ **A.** 1050GB (approximately 1TB) total; either a 2TB local server HDD for cost efficiency or cloud storage for easier access from any school computer would be appropriate

○ **B.** 500GB total; a single external HDD would be sufficient

○ **C.** 50GB total; a USB flash drive shared between students

○ **D.** The calculation is impossible without knowing the file types

Q29 (3 marks)

A computer has 8GB of RAM and a user opens several large applications until memory usage reaches 95%. They then try to open another application. Describe what happens and why the system becomes slow.

○ **A.** The computer immediately crashes because RAM cannot exceed 100%

○ **B.** The computer automatically purchases more RAM from the cloud

○ **C.** The new application simply fails to open and displays an error

○ **D.** The operating system uses virtual memory, moving less-active data to the hard drive to make room; performance degrades because the system must frequently swap data between fast RAM and slow storage

Q30 (3 marks)

Compare the internal operation of SSDs and HDDs, explaining how their different technologies affect performance, durability, and power consumption.

○ **A.** SSDs and HDDs operate identically; the only difference is marketing

○ **B.** HDDs use electronic circuits while SSDs use magnetic platters; this makes HDDs faster and more durable

○ **C.** SSDs store data in flash memory chips using electrical charges with no moving parts, resulting in faster access, greater shock resistance, and lower power consumption; HDDs use spinning magnetic platters with moving read/write heads, offering lower cost per GB but slower access, mechanical vulnerability, and higher power use

○ **D.** SSDs have moving parts that are smaller than HDD moving parts, which is why they are faster

Total: **55** marks • Time: **30 min**

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